

CSE525 Grad. Algo.

Monsoon 2025, Lecture 15

QUIZ

MakeSet(1)

MakeSet(2)

MakeSet(3)

MakeSet(4)

MakeSet(5)

MakeSet(6)

Union(3,4)

Union(1,2)

Union(6,5)

Union(3,5)

A. Number of sets ?

B. Is Find(6)=Find(1) ?

C. What is parent(5)?

1 / 2 / 3 / 4 / 5 / 6

D. For which x is parent(x) = x ?

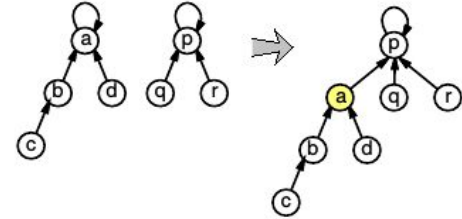
1 / 2 / 3 / 4 / 5 / 6

E. Is Find(3)=parent(6) ?

```
MAKESET(x):  
  parent(x) ← x
```

```
FIND(x):  
  while x ≠ parent(x)  
    x ← parent(x)  
  return x
```

```
UNION(x, y):  
   $\bar{x} \leftarrow \text{FIND}(x)$   
   $\bar{y} \leftarrow \text{FIND}(y)$   
  parent( $\bar{y}$ ) ←  $\bar{x}$ 
```



n : total number of elements at the
time of invocation

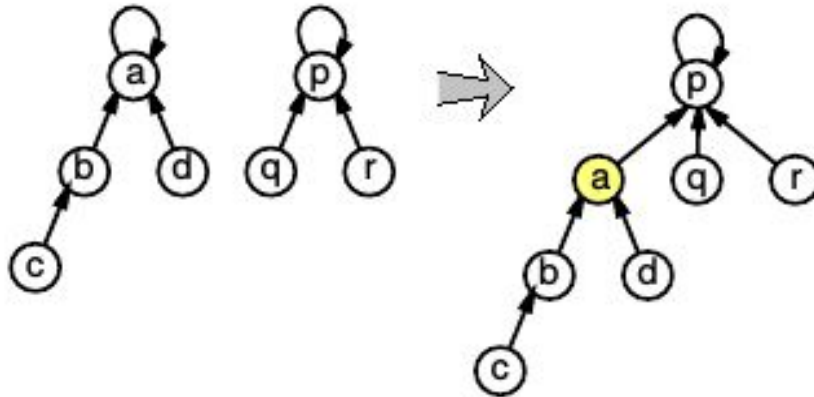
Disjoint-Set using reversed trees

MAKESET(x):
 $\text{parent}(x) \leftarrow x$

FIND(x):
while $x \neq \text{parent}(x)$
 $x \leftarrow \text{parent}(x)$
return x

UNION(x, y):
 $\bar{x} \leftarrow \text{FIND}(x)$
 $\bar{y} \leftarrow \text{FIND}(y)$
 $\text{parent}(\bar{y}) \leftarrow \bar{x}$

Second loses
leadership.



Tight complexity of ...

- Makeset
- Find
- Union

Give a **technique** to
generate a sequence of ops.
on n elements so that
 $\text{Find}(x)$ takes $\Theta(n)$ time.

Tight Example for Find() Worst-case Complexity.

(i) Complexity is $O(n)$: If there are n elements in the universe, any path can have at most $O(n)$ nodes to the leader. Since Find() involves traversing parent pointers, complexity of any Find() is $O(n)$.

(ii) Complexity is $\Omega(n)$: A sequence of operations on n elements ending with a Find(x) such that the Find(x) step takes $\Omega(n)$ steps.

```
def generate_worstcase(n):  
1.     for i=1...n: Makeset(i);  
2.     for i=n-1...1: Union(i,n);  
3.     Find(n);
```

The reversed tree after line #2 will be a single chain rooted at 1 and ending at n . Thus, the Find(n) step in line #3 will require $\Omega(n)$ steps.

Reversed-Trees + Union-by-depth

During union, join the shorter tree to the root of the deeper tree (not the other way).

Q: How to know “shorter” and “deeper” ? How to “join” ?

```
MAKESET(x):  
  parent(x) ← x  
  depth(x) ← 0
```

```
FIND(x):  
  while x ≠ parent(x)  
    x ← parent(x)  
  return x
```

```
UNION(x, y)  
   $\bar{x} \leftarrow \text{FIND}(x)$   
   $\bar{y} \leftarrow \text{FIND}(y)$   
  if  $\text{depth}(\bar{x}) > \text{depth}(\bar{y})$   
     $\text{parent}(\bar{y}) \leftarrow \bar{x}$   
  else  
     $\text{parent}(\bar{x}) \leftarrow \bar{y}$   
    if  $\text{depth}(\bar{x}) = \text{depth}(\bar{y})$   
       $\text{depth}(\bar{y}) \leftarrow \text{depth}(\bar{y}) + 1$ 
```

If depths are same, first leader loses leadership and second leader becomes the new leader.

Try: Union(1,2) Union(3,4) Union(2,4) Union(5,6) Union(4,6) Union(7,8) Union(9,10) Union(8,10) Union(6,10)

Reversed-Trees + Union-by-depth

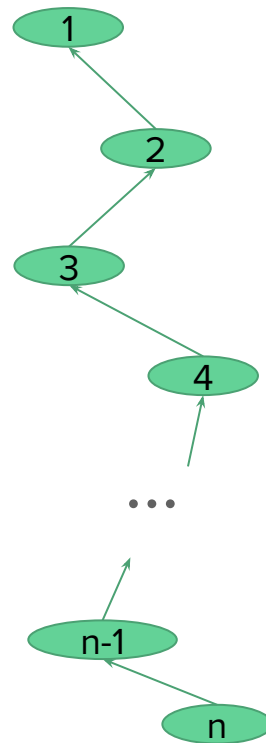
```
FIND(x):  
  while  $x \neq \text{parent}(x)$   
     $x \leftarrow \text{parent}(x)$   
  return  $x$ 
```

If depths are same, second leader becomes new leader.

```
UNION(x, y)  
   $\bar{x} \leftarrow \text{FIND}(x)$   
   $\bar{y} \leftarrow \text{FIND}(y)$   
  if  $\text{depth}(\bar{x}) > \text{depth}(\bar{y})$   
     $\text{parent}(\bar{y}) \leftarrow \bar{x}$   
  else  
     $\text{parent}(\bar{x}) \leftarrow \bar{y}$   
    if  $\text{depth}(\bar{x}) = \text{depth}(\bar{y})$   
       $\text{depth}(\bar{y}) \leftarrow \text{depth}(\bar{y}) + 1$ 
```

Claim: Worst-case complexity of Find() is $\Theta(n)$.

Proof: We have to prove both (a) complexity is $O(n)$, (b) complexity is $\Omega(n)$.



Reversed-Trees + Union-by-depth

Lemma: For any set A , $|A| \geq 2^{\text{depth}(\text{leader}(A))}$.

Proof: Let x = leader of A . Induction on the depth of the set with x as leader.

- Base case: $\text{depth}(x) = 0$: show that $\text{size}(x\text{'s set}) \geq 1$
- Fix a general d . Assume that $|x\text{'s set}| \geq 2^{\text{depth}(x)}$ as long as $\text{depth}(x) \leq d$
- Consider a scenario where $\text{depth}(x) = d+1$. Prove that $|x\text{'s set}| \geq 2^{d+1}$.
Analyse the operation which changed the depth of x 's set to $d+1$.

Reversed-Trees + Union-by-depth

During union, join the shorter tree to the root of the deeper tree (not the other way).

Lemma: For any set A , $|A| \geq 2^{\text{depth}(\text{leader}(A))}$.

This implies: $\text{depth}(\text{leader}(A)) \leq \log(\# \text{elements of } A)$

Q: How are the complexities affected?

- ... of MakeSet
- ... of Find
- ... of Union

Is the analysis tight?

Can the operations be improved to $O(1)$ using better analysis?

QUIZ

Suppose x is the leader of a set A at any point in time. Select the tightest bounds for each question.

A. If A has 34 elements, what can be said about $\text{depth}(x)$?

- | | | |
|-----------------------------|-----------------------------|--------------------------|
| a. $\text{depth}(x) \geq 5$ | b. $\text{depth}(x) > 5$ | c. $\text{depth}(x) = 5$ |
| d. $\text{depth}(x) < 5$ | e. $\text{depth}(x) \leq 5$ | f. depends on A |

B. Let y be another element of A at that time. What can be said about $\text{depth}(y)$?

- | | | |
|-----------------------------|-----------------------------|--------------------------|
| a. $\text{depth}(y) \geq 5$ | b. $\text{depth}(y) > 5$ | c. $\text{depth}(y) = 5$ |
| d. $\text{depth}(y) < 5$ | e. $\text{depth}(y) \leq 5$ | f. depends on A |

C. Several more union calls has happened and x is the still the leader. After one more $\text{union}()$, A now has 64 elements, and x is no longer the leader of A . What can be said about $\text{depth}(x)$?

- | | | |
|-----------------------------|--------------------------|-----------------------------|
| a. $\text{depth}(x) \leq 5$ | b. $\text{depth}(x) = 5$ | c. $\text{depth}(x) \geq 5$ |
| d. $\text{depth}(x) \leq 6$ | e. $\text{depth}(x) = 6$ | f. $\text{depth}(x) \geq 6$ |

Quiz

1. After line 5, what does Find(1) return?
2. After line 5, what is depth(1)?
3. After line 5, what is depth(Find(1))?
4. After line 7, what is depth(1)?
5. After line 7, what is depth(Find(1))?
6. After line 20 (which is not shown), what is depth(parent(1))?
7. After line 20 (which is not shown), if the set containing 1 has 65 elements, what is depth(Find(1))? Select all the possible values.
8. Consider the path from 1 to its leader L:
 $1 \rightarrow a \rightarrow b \rightarrow c \rightarrow \dots \rightarrow L$
Let x be any node on this path. Fill in the blanks.
depth(x) _____ depth(x)

MakeSet(1)

...

MakeSet(n)

1. U(1,2)

2. U(3,4)

3. U(5,6)

4. U(7,8)

5. U(1,3)

6. U(5,7)

7. U(1,7)

...

Tight bounds on Find(...)

Exercise!

MakeSet(1)

...

MakeSet(n)

U(1,2)

U(3,4)

U(1,3)

U(5,6)

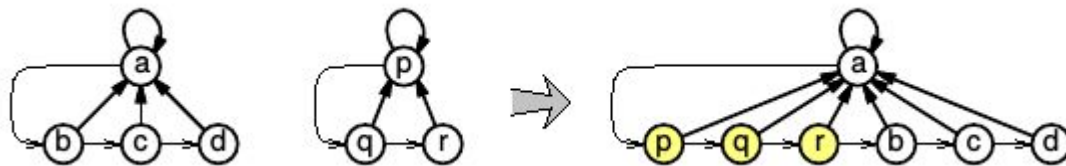
U(7,8)

U(5,7)

U(1,7)

...

Shallow Reversed Trees + Threading



First leader becomes
new leader

```
MAKESET(x):
  leader(x) ← x
  next(x) ← Null
```

```
FIND(x):
  return leader(x)
```

```
UNION(x, y):
   $\bar{x} \leftarrow \text{FIND}(x)$ 
   $\bar{y} \leftarrow \text{FIND}(y)$ 
   $y \leftarrow \bar{y}$ 
  leader(y) ←  $\bar{x}$ 
  while (next(y) ≠ NULL)
     $y \leftarrow \text{next}(y)$ 
    leader(y) ←  $\bar{x}$ 
  next(y) ← next( $\bar{x}$ )
  next( $\bar{x}$ ) ←  $\bar{y}$ 
```

Complexity of ...

- Makeset
- Find
- Union (# leader changes)

Tight bound on Worst-case Complexity of Union

Give a technique to **generate** a sequence of ops. on **n** elements so that Union(x,y) takes $\Theta(n)$

U(1,2)

U(3,4)

U(1,4)

U(5,6)

U(7,8)

U(5,8)

U(1,8)

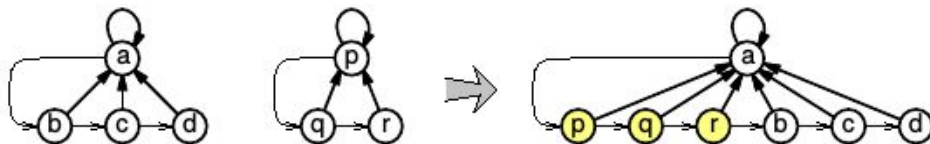
...

U(1,16)

...

U(1,n)

Shallow Reversed Trees + Threading + Wtd Union



During union, join the smaller tree to the root of the larger tree.

MAKEWEIGHTEDSET(x):

$\text{leader}(x) \leftarrow x$

$\text{next}(x) \leftarrow \text{Null}$

$\text{size}(x) \leftarrow 1$

FIND(x):

return $\text{leader}(x)$

UNION(x, y):

$\bar{x} \leftarrow \text{FIND}(x)$

$\bar{y} \leftarrow \text{FIND}(y)$

$y \leftarrow \bar{y}$

$\text{leader}(y) \leftarrow \bar{x}$

while ($\text{next}(y) \neq \text{Null}$)

$y \leftarrow \text{next}(y)$

$\text{leader}(y) \leftarrow \bar{x}$

$\text{next}(y) \leftarrow \text{next}(\bar{x})$

$\text{next}(\bar{x}) \leftarrow \bar{y}$

WEIGHTEDUNION(x, y)

$\bar{x} \leftarrow \text{FIND}(x)$

$\bar{y} \leftarrow \text{FIND}(y)$

if $\text{size}(\bar{x}) > \text{size}(\bar{y})$

$\text{UNION}(\bar{x}, \bar{y})$

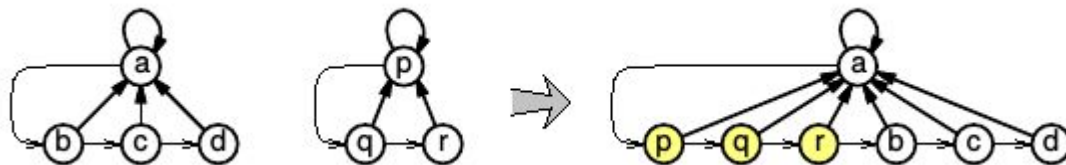
$\text{size}(\bar{x}) \leftarrow \text{size}(\bar{x}) + \text{size}(\bar{y})$

else

$\text{UNION}(\bar{y}, \bar{x})$

$\text{size}(\bar{y}) \leftarrow \text{size}(\bar{x}) + \text{size}(\bar{y})$

Shallow Reversed Trees + Threading + Wtd Union



Worst-case complexity of ... (on n -element sets)

- MakeSet
- Find
- WtdUnion

Analyse the earlier “bad” example.

Tight bounds on $\text{WtdUnion}(\dots)$

Exercise!

$U(1,2)$

$U(3,4)$

$U(1,4)$

$U(5,6)$

$U(7,8)$

$U(5,8)$

$U(1,8)$

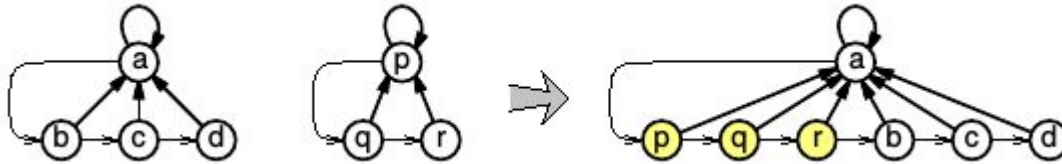
...

$U(1,16)$

...

$U(1,n)$

Ammortized Analysis of WtdUnion



How often can the worst-case situation happen?

Complexity $\approx$ change of leader pointer

Lemma: If x belongs to a set with n elements, then x 's leader has changed $O(\log(n))$ times.

WEIGHTEDUNION(x, y)

$\bar{x} \leftarrow \text{FIND}(x)$

$\bar{y} \leftarrow \text{FIND}(y)$

if $\text{size}(\bar{x}) > \text{size}(\bar{y})$

$\text{UNION}(\bar{x}, \bar{y})$

$\text{size}(\bar{x}) \leftarrow \text{size}(\bar{x}) + \text{size}(\bar{y})$

else

$\text{UNION}(\bar{y}, \bar{x})$

$\text{size}(\bar{y}) \leftarrow \text{size}(\bar{x}) + \text{size}(\bar{y})$

UNION(x, y):

$\bar{x} \leftarrow \text{FIND}(x)$

$\bar{y} \leftarrow \text{FIND}(y)$

$y \leftarrow \bar{y}$

$\text{leader}(y) \leftarrow \bar{x}$

 while ($\text{next}(y) \neq \text{NULL}$)

$y \leftarrow \text{next}(y)$

$\text{leader}(y) \leftarrow \bar{x}$

$\text{next}(y) \leftarrow \text{next}(\bar{x})$

$\text{next}(\bar{x}) \leftarrow \bar{y}$

QUIZ

Consider a sequence of 10 MakeSet calls (so, universe has 10 elements). After this several Union(...) calls are made.

1. Number of sets after 1 union call =
2. Number of sets after 2 union calls =
3. Number of sets after 7 union calls =
4. Total number of elements involved in the 2nd union call :
5. Total number of elements involved in the 3rd union call :